Xiaowen Zhang

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Research Interests

- **Cosmology:** large-scale structure and nonlinear structure formation, N-body and hydrodynamic simulations, emulators and surrogate models, simulation-based inference
- **Machine learning:** generative models, graph neural networks, representation learning and foundation models for simulation data, LLM agents for simulation workflows
- N-body simulations (1083); Large-scale structure of the universe (902); Neural networks (1933)

Education

- **Carnegie Mellon University** — *Sep. 2021 - Present*
  Ph.D. in Physics, expected 2027 (Advisors: Prof. Tiziana Di Matteo and Prof. Rupert Croft) — Pittsburgh, USA
- **ShanghaiTech University** — *Sep. 2016 - Jun. 2020*
  B.S. in Physics — Shanghai, China
- **University of California, Berkeley** — *Aug. 2018 - May 2019*
  Visiting Student, Berkeley Physics International Education (BPIE) Program — Berkeley, USA

Research Experience

- **McWilliams Center for Cosmology & Astrophysics** — *Sep. 2021 - Present*
  Graduate Research Assistant — Carnegie Mellon University
  - Super-resolution of N-body simulations across cosmic time: 512× more particles, validated on power spectra, halo mass functions, and merger trees. First author, *MNRAS* [6].
  - Deterministic emulator that recovers the specific realization implied by a simulation's initial conditions (first author, *OJAp* [5]); the same method, applied by a mentee to the Lyman-α forest, reaches sub-percent accuracy at 260× the speed of the hydrodynamic solver (second author [4]).
  - Cosmology-aware super-resolution: one model for every cosmology in the Quijote suite. First author, submitted [1].
  - SimAgents, a multi-agent LLM system that configures and runs MP-GADGET simulations from the literature: 80% fewer setup errors, 83× faster setup. First author [2].
- **Institute of Mathematical Sciences** — *Aug. 2020 - Aug. 2021*
  Research Assistant — ShanghaiTech University
  - Topological data analysis of high-dimensional cosmological datasets; tropical-geometry representations of neural network decision boundaries.

Publications

[1] **Zhang, X.**, Lachance, P., Croft, R. A. C., Di Matteo, T., & Villaescusa-Navarro, F. (2026). *AI-assisted super-resolution cosmological simulations V: Cosmology-aware super-resolution*. Submitted.

[2] **Zhang, X.**, Bi, Z., LaChance, P., Wang, X., Di Matteo, T., & Croft, R. (2025). *SimAgents: Bridging Literature and the Universe Via A Multi-Agent Large Language Model System*. *Proc. IJCNLP-AACL 2025: System Demonstrations*, 55–66. Association for Computational Linguistics.

[3] Zhou, Y., Di Matteo, T., Bird, S., Croft, R. A. C., Ni, Y., Yang, Y., Chen, N., Lachance, P., **Zhang, X.**, & Hafezianzadeh, F.*. (2025). *The ASTRID Simulation at $z = 0$: from Massive Black Holes to Large-scale Structure*. arXiv:2510.13976.

[4] Hafezianzadeh, F.*, **Zhang, X.**, Ni, Y., Croft, R. A. C., Di Matteo, T., Qezlou, M., & Bird, S. (2025). *An AI super-resolution field emulator for cosmological hydrodynamics: the Lyman-α forest*. arXiv:2507.16189.

[5] **Zhang, X.**, Lachance, P., Dasgupta, A.*, Croft, R., Di Matteo, T., Ni, Y., Bird, S., & Li, Y. (2025). *AI-assisted super-resolution cosmological simulations IV: An emulator for deterministic realizations*. *The Open Journal of Astrophysics*, **8**, E13.

[6] **Zhang, X.**, Lachance, P., Ni, Y., Li, Y., Croft, R., Di Matteo, T., Bird, S., Feng, Y., et al. (2024). *AI-assisted super-resolution cosmological simulations III: time evolution*. *MNRAS*, **528**, 281–293.

**Denotes mentored student.*

Talks & Presentations

- **KAAI Workshop on Foundation Models & Agentic Methods for Astrophysical Simulation** *Aug. 2026*
  Talk: *A Shared "Language" for Simulated Universes: Foundation Representations for Galaxy Catalogs* Pittsburgh. PA
- **2nd Annual Cosmic Horizons Conference (CosmicAI)** *Jul. 2026*
  Talk: *AI Accelerators for Cosmological Simulations: Super-Resolution and Emulation* Charlottesville, VA
- **COSMO-25: 28th Int. Conference on Particle Physics & Cosmology** *Oct. 2025*
  Talk: *Accelerating cosmological simulations with super-resolution; LLMs for running simulations* Carnegie Mellon University
- **Machine Learning in Astronomical Surveys** *Nov. 2023*
  Poster: *AI-assisted Super-Resolution Cosmological Simulations IV: An Emulator for Deterministic Realizations* Flatiron Institute
- **Cosmic Connections: A ML × Astrophysics Symposium** *May 2023*
  Poster: *AI-assisted Super-Resolution Cosmological Simulations III: Time Evolution* Flatiron Institute

Computing Allocations

- **NSF ACCESS Allocation** *2025-*
  Learning Cosmological Simulations with Graph Neural Networks **PI**, PHY250036
- **NSF ACCESS Allocation** *2024-2025*
  An Emulator for Transforming Random Super-Resolution Realizations to Deterministic High-Resolution Outputs PHY240032

Research Support & Collaborations

- **KAAI Program, McWilliams Center, Carnegie Mellon University** *2026*
  Interdisciplinary program on AI for astrophysical simulation; summer research support
- **Block Center Seed Fund, *AI Acceleration of Scientific Research***
  Team member (PI: Prof. Rupert Croft); supported the SimAgents project Carnegie Mellon University
- **Travel Support, Simons Foundation / Flatiron Institute** *2023*
  Cosmic Connections: A ML × Astrophysics Symposium

Teaching & Mentoring

- **Research Mentees**
  - Fatemeh Hafezianzadeh (graduate), *Super-resolution of Lyman-α Forest Simulations*, 2024-present — first author of publication [4].
  - Ankita Dasgupta (undergraduate), *An Emulator for Deterministic Super-Resolution Output*, 2023-2024 — co-author of publication [5].
  - Aidan Catalano & Matthew Pan (undergraduates), *Learning Cosmological Simulations with GNNs*, 2025-present.
- **Graduate Teaching Assistant, Carnegie Mellon University**
  - *33-152 Matter & Interactions II*, *33-142/122 Introductory Physics II*, *33-231 Physical Analysis* (recitations, office hours, grading); *33-341 Thermal Physics I* (grader); *33-101 First Year Seminar* (instructor: Prof. Tiziana Di Matteo) — designed and taught the module on using LLM tools in coursework, and guest-lectured.

Service & Organization

- **KAAI Workshop: Foundation Models & Agentic Methods for Astrophysical Simulation** *Aug. 31 - Sep. 3, 2026*
  Hackathon track lead, Robust Cosmological Inference*; designed the task and its public toolkit* Carnegie Mellon University

Open Source Software

- **CosmoSR**: Generative Adversarial Network that super-resolve Quijote Simulations at multiple cosmologies, used in publication [1]
- **SimAgents**: Multi-agent LLM framework that turns a physical setup described in the literature into a runnable MP-GADGET configuration and first-look analysis, with a validated database of simulation parameters from published suites for evaluation; used in publication [2].
- **SREmulator**: Neural models that transform stochastic super-resolution outputs into deterministic realizations matching specific initial conditions; used in publication [5].
- **map2map (stylegan2)**: Style-conditioned super-resolution models for N-body simulations (512× mass resolution), developed on Yin Li's `map2map` framework — upstream PR merged, 2021; used in publications [5] and [6].

Technical Skills

- **Simulations:** GADGET, MP-GADGET; Quijote, CAMELS, and ASTRID data products; halo finding and merger trees
- **Computing:** Python (PyTorch, JAX), CUDA, multi-node multi-GPU training on national HPC systems

Languages

- Mandarin Chinese (native); English (fluent).